Theme Warm

Optimization

Advanced Statistics Lab

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Formal Definition

Choose values that make a function small or large.

We start with an ordinary function.

Least squares and likelihood come later.

Theme

Basic Optimization Problem

OBJECTIVE

Let θ be the parameter vector and Θ the allowed parameter space.

$$\hat{\theta} \in \arg\min_{\theta \in \Theta} f(\theta)$$

For maximization:

$$\arg\max_{\theta \in \Theta} f(\theta) = \arg\min_{\theta \in \Theta} - f(\theta)$$

Theme

Symbol Map

Symbol	Meaning
θ	Candidate value or parameter vector.
Θ	Allowed set of candidate values.
$\hat{\theta}$	Optimizing value, marked with a hat because it has been selected or estimated.
$f(\theta)$	Objective function.
$\arg\min$	Input value that gives the minimum.
$\arg\max$	Input value that gives the maximum.

Theme

Constrained Problems

CONSTRAINTS

$$\hat{\theta} \in \arg\min_{\theta \in \Theta} f(\theta)$$

$$\text{subject to } g_j(\theta) \le 0, \quad j = 1, \ldots, m$$

$$h_k(\theta) = 0, \quad k = 1, \ldots, r$$

Constraints narrow the parameter space.

Symbol	Meaning
$g_j(\theta)$	The jth inequality constraint.
$h_k(\theta)$	The kth equality constraint.
m	Number of inequality constraints.
r	Number of equality constraints.

Theme

The Workflow

Model | Objective | Search | Constraints

1. Define the model.
2. Define the objective.
3. Search for good parameter values.
4. Check whether the result makes statistical and biological sense.

First Example: Fence A Paddock

MODEL

80 meters of fence.

$$2w + 2\ell = 80$$

Divide by 2:

$$w + \ell = 40$$

So:

$$\ell(w) = 40 - w$$

$$A(w) = w(40 - w)$$

Theme

Area Function

OBJECTIVE

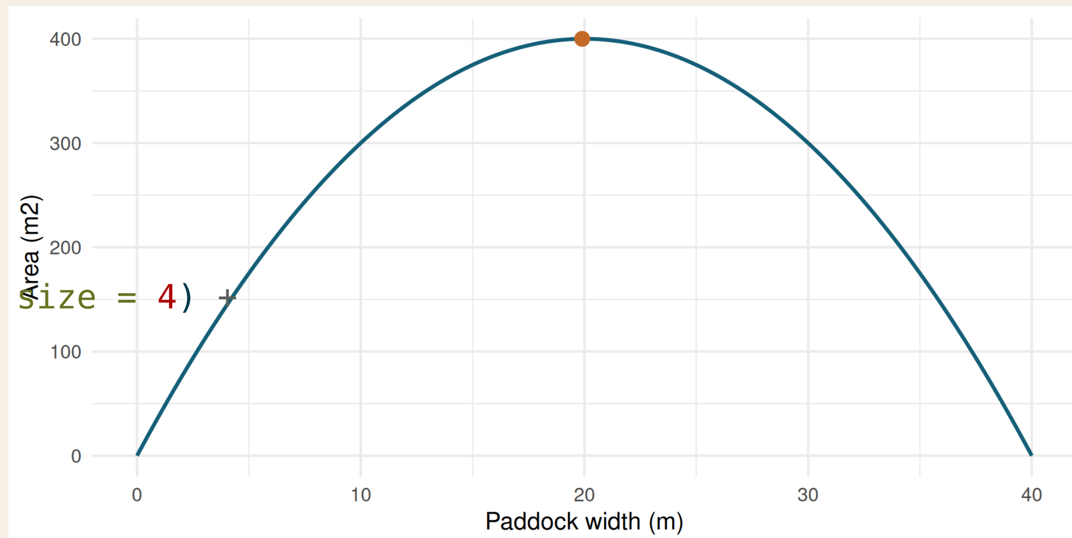
```r
fence_m <- 80

paddock_area <- function(width, fence_m = 80) {
  length <- fence_m / 2 - width
  width * length
}

candidate_widths <- seq(0, fence_m / 2, length.out = 200)
area_grid <- data.frame(width = candidate_widths)
area_grid$area <- paddock_area(area_grid$width, fence_m)
best_grid <- area_grid[which.max(area_grid$area), ]
```

$$\widehat{w} = \arg\max_{0 \le w \le 40} A(w)$$

Theme

Objective Shape

```r
ggplot(area_grid, aes(width, area)) +
  geom_line(color = "#155f7a", linewidth = 1.2) +
  geom_point(data = best_grid, color = "#c46a2b", size = 4) +
  labs(x = "Paddock width (m)", y = "Area (m2)")
```



Search With optim()

SEARCH

Maximize area by minimizing negative area.

```r
negative_area <- function(width, fence_m = 80) {
  -paddock_area(width, fence_m)
}

fit_area <- optim(
  par = 10,
  fn = negative_area,
  fence_m = fence_m,
  method = "L-BFGS-B",
  lower = 0,
  upper = fence_m / 2
)

data.frame(width = fit_area$par, area = -fit_area$value)
```

```
  width area
1    20  400
```

Lab notes

Theme

Bound Constraints

CONSTRAINTS

$$0 \le w \le 40$$

```r
fit_tight_area <- optim(
  par = 10,
  fn = negative_area,
  fence_m = fence_m,
  method = "L-BFGS-B",
  lower = 0,
  upper = 15
)

data.frame(
  problem = c("correct bounds", "too-tight upper bound"),
  width = c(fit_area$par, fit_tight_area$par),
  area = c(-fit_area$value, -fit_tight_area$value)
)
```

```
                problem width area
1        correct bounds    20  400
2 too-tight upper bound    15  375
```

Lab notes

Where This Goes Next

Theme

MODEL

Same workflow, statistical objectives:

Topic	Objective
Least squares	Minimize squared vertical distances.
Maximum likelihood	Maximize probability of the observed data.
Animal models	Balance fit and shrinkage.

Exercises

1. Change the amount of fencing.
2. Change the starting value in `optim()`.
3. Make the upper bound too tight.
4. Rewrite the objective for a three-sided fence.